DAEN/ISEN 427 Homework 1

Due date: Tue Sep 9, 2025 (before class)

Rubric: Full credit may be given for any reasonable Bayesian analysis.

1. Explain the difference between a *confounder* and a *collider* in the context of a DAG. Why is it important not to condition on a collider when trying to estimate a causal effect?

   Answer:

2. You are studying whether a new training program (A) improves job performance (Y). Based on domain knowledge, age and prior experience influence both program participation and job performance. Gender is believed to influence only program participation. Construct a DAG using this information, and identify the minimum set of covariates you would need to adjust for to estimate the causal effect of A on Y.

   Answer:

3. Consider a DAG where a treatment variable A causally affects an outcome Y, and there exists a confounder C that influences both A and Y.

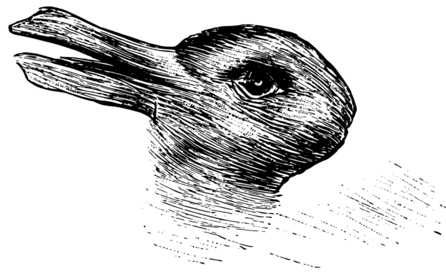
   (a) Explain why the conditional probability $P(Y \mid A)$ is not equal to the causal effect $P(Y \mid \text{do}(A))$.

   (b) Describe how adjusting for C allows us to estimate the causal effect using observational data.

   Note: the *do* operator means we are intervening to set A to a value rather than just observing it.

   Answer:

4. A research article entitled "The Easter bunny in October: Is it disguised as a duck?" explained that "Very little is known about the looks of the Easter bunny on his non-working days." To investigate, the authors showed an "ambiguous drawing of a duck/rabbit...to...265 subjects on Easter Sunday and to 276 different subjects on a Sunday in October of the same year." The authors report: "Whereas on Easter the drawing was significantly more often recognized as a bunny, in October it was considered a bird by most subjects." The drawing shown by the authors in their study was similar to the following figure. Provide a Bayesian perceptual explanation for the authors' results.



Answer:

5. Suppose that 5% of men and 0.25% of women are color-blind. A person is chosen at random and that person is color-blind. What is the probability that the person is male? (Assume males and females to be in equal numbers.)

Answer:

6. Suppose the state of the world s can take the values in $\{s_1, s_2, \ldots, s_n\}$. Let x denote an observation and assume $P(x|s = s_i) = P(x|s = s_j) \quad i \neq j$, i.e. the likelihood of observation x is the same for all states of the world. Prove using Bayes' rule the posterior distribution is identical to the prior distribution.

Answer: